



## DriftTeR: A Drone Identification Technique using RF signals

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## ABSTRACT

The civilian drone market is experiencing explosive growth, with projections estimating it will hit USD 54.81 billion by 2030. This surge in drone numbers brings with it significant privacy and security challenges. To defend critical infrastructure and safeguard personal privacy from misuse, an effective drone detection system has become essential. There is a demand for detection solution that is not only efficient and accurate but also robust, cost-effective, and scalable to meet the evolving needs of this rapidly expanding field. In this paper, we present DriftTeR, a drone detection, identification and classification model based on the radio frequency signals. Firstly we employ wavelet domain extraction and 3-stage wavelet decomposition during RF signal preprocessing. Secondly, we employ traditional machine learning, deep learning and ensemble learning models to evaluate effectiveness. Thirdly, we evaluate performance of DriftTeR against drone detection, drone manufacturer identification and drone model identification. The performance of the approach is evaluated against benchmark dataset and is found to be effective and accurate.

## 1. Introduction

Drones or Unmanned Aerial Vehicles (UAVs) are remotely controlled and commanded quadcopters / small aircrafts used for defense to rescue operations. Although drones were used chiefly for military objectives, their usage for civilian applications has increased dramatically in recent years. The dominance of the digital lifestyle has resulted in the employment of drones for applications such as patrolling and policing; search and rescue; agrotech and videography; and preventing and identifying poachers. Drones have also been detected in violations of non-flying territories, resulting in a considerable risk that terrorists may utilize them to inflict damages (Elands et al., 2016). The global drone market is estimated to surpass 42.8 billion dollars by 2025, with an annual increase of 13.8% (Fu et al., 2021).

The popularity and affordability of drones have also increased their use in criminal operations. The drone seized from the crime scene, as well as the devices to which it was connected, is critical for cyber forensics investigation. Pieces of evidence such as a drone ID, previous flight locations, camera shots, logs, and software used can be extracted from a drone and its ground controller. Forensic investigation of a drone is challenging since it relies on volatile memory and dynamic data, which

can be lost upon battery discharge (Bouafif et al., 2018; Sihag et al., 2023).

Drone identification is a subdomain of drone forensics that detects and identifies drones based on their traits (for example, optical data, radio frequency, and radar signals). Radio frequency (RF) fingerprinting relies on the idea that each device has distinct traits that distinguish it (Dave et al., 2022). Different drone detection methods have been introduced based on traits such as sound, vision, radar, and RF signals. RF based drone fingerprinting is a technique for identifying individual UAV based on their unique RF signature. This is accomplished by analyzing the RF signals sent by the drone and creating a fingerprint or signature of its specific RF characteristics. The RF signature includes the particular frequency and waveform characteristics of the drone's transmission, which can be used to differentiate it from other devices. It also helps to identify the manufacturer and model of the drone and distinguish it from others by utilizing its unique characteristics. Conventional detection systems (radar, video, acoustic) may not effectively detect drones due to various challenges like small size, resemblance to birds, lack of line of sight, weather, and environmental noise. RF-based detection, however, is more effective as it can detect drones from a distance, regardless

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of size, weather, and lighting conditions, even in beyond-line-of-sight (BLOS) conditions (Basak et al., 2021a; Choudhary et al., 2024).

The literature has proposed RF-based techniques for drone detection (Al-Sa'd et al., 2019; Kosolyudthasarn et al., 2018; Nguyen et al., 2016). In Nguyen et al. (2016) a joint active and passive system was proposed that uses passive RF sensing at 2.4 GHz to detect drones based on timing information of message communication to the Ground Control Station (GCS). In Kosolyudthasarn et al. (2018), the authors used a packet length signature for detection. For drone identification, conventional methods based on fingerprinted features (using modulation, statistical, transient and wavelet methods) have been used in the literature. It is challenging to distinguish and categorize the signal from a drone, since wireless devices operating in ISM bands, such as Bluetooth, ZigBee, and Wifi, employ the same transmission mechanism (Direct Sequence Spread Spectrum and Frequency Hopping Spread Spectrum). This encourages us to thoroughly investigate the categorization of drone signals on a larger dataset with more commercial drones. We provide an RF-based drone classification approach in this work. Following are the key contributions of this paper:

- DrIfTeR, a drone RF fingerprinting model based on ensemble learning, is proposed by utilizing ML algorithms to detect, identify, and classify drones using RF fingerprints.
- The performance of DrIfTeR for drone detection and multi-drone classification is evaluated against benchmark datasets.
- The proposed approach can detect, identify, and classify drones with high accuracy. In terms of accuracy, F1 score, and other characteristics, the model outperforms existing popular drone RF fingerprinting approaches.

**Structure of the paper:** In the following section, we discuss the related work available in the literature. The background of drone architecture and its communication is discussed in Section 3. This work introduces DrIfTeR, a drone forensic model for drone detection, identification, and classification, in Section 4. Section 5 includes a detailed discussion about the modes of evaluation used for evaluation and the performance of DrIfTeR with various classification algorithms on the given dataset. Furthermore, Section 6 gives the conclusion about this study and the related future work.

## 2. Related work

Drone radio fingerprinting is an emerging area of research that focuses on using the unique characteristics of drone radio signals to identify and track drones. Several studies have explored the use of RF signal analysis to extract features from drone radio transmissions. Machine learning algorithms such as Support Vector Machines (SVMs), random forests, and deep neural networks have been employed to classify drone radio fingerprints and distinguish them from other RF signals. Other studies have also investigated the effectiveness of using multiple sensors and modalities, such as acoustic and video data, to enhance drone detection and identification. In this section, we present the existing contributions to the problem.

Clemente et al. (2021) introduced a method for micro-Doppler identification and drone fingerprinting that relies on Chebychev moments-based features. This approach offers flexibility and requires minimal computing costs. The effectiveness of the proposed framework was evaluated at different stages, demonstrating its ability to distinguish between drones. Al-Emadi et al. (2019) explored the use of deep learning algorithms such as a convolutional neural network (CNN), Recurrent Neural Network (RNN), and a Convolutional Recurrent Neural Networks (CRNN) for detecting and identifying drones using audio recordings of drone activity. It evaluated the performance of different neural networks using its dataset. The research also tested the effectiveness in real-world scenarios. A technique for detecting a drone's state by using the communication traffic shared between a drone and its controllers is offered in Sciancalepore et al. (2019) by Sciancalepore et al. The presented

method is entirely passive and relies on low-cost general-purpose hardware. Real-time communication from a drone running the widely used ArduCopter was captured and analyzed to determine the drone's state.

Picking a Needle in a Haystack (PiNcH) (Sciancalepore et al., 2020) is designed to analyze eavesdropped communications using common categorization techniques, including packet inter-arrival timing and length. It is entirely passive and uses inexpensive public network infrastructure. Valid communication traces generated by a drone using the widely used ArduCopter open-source firmware have been utilized for testing and evaluating the performance as well and various publicly accessible wireless traces have also been used. It demonstrates that network traffic analysis may be used efficiently for drone recognition and status differentiation, paving the path for further study in the field. The primary goal of Sheikh et al. (2019) is to offer a recommended collection of characteristics from a wireless network to be used as predictive variables in the categorization of flying drones for various ML algorithms. Moreover, it also gives a comparison of the performance of four models, namely Decision Tree, Logistic Regression, Discriminant Analysis, and k-nearest neighbors (KNN), in regards to recognizing the flying drones. Paper Xu et al. (2020) proposes an RF signal detection approach for detecting numerous drone intrusions. The time-frequency-energy distribution properties of a single transient control, as well as video signals, are obtained using the Short-Time Fourier Transform. PCA is used in the signal transformation to lower the dimensions of the RF feature set. The RF signal feature data is employed and trained using the SVMs and KNN algorithms for detecting the existence and quantity of invading drones. Authors in Elyoussef and Altamimi (2021) conduct numerous experiments to improve the existing knowledge on detecting and categorizing drones. The findings indicate that image classifiers may perform comparably well, if not better than, 1D coefficient-based classifiers. It moreover shows the influence of multiple Discrete Fourier Transform averaging techniques, which are commonly used to minimize the noise variation of power spectra estimations. It also shows that raising the average while making the photos improves accuracy, thereby decreasing training costs (Table 1).

Drones have numerous uses, including drone monitoring devices, anti-drone systems, intrusion detection, military, space programs, shipping, agribusiness, search and rescue, and online media operators. In sensitive cases, such as infiltration, it is difficult to pinpoint a particular drone and its kind. A drone classification approach utilizing fixed boundary empirical wavelet sub-bands of RF signature and CNN is presented in Bremnes et al. (2022). The decomposing RF signatures are obtained from drone controllers using 16 fixed boundary empirical wavelet sub-band signals. The sub-band signals are then put into a light CNN algorithm, which is used to categorize multiple kinds of drones. In consideration of forensics objectives, authors in Li et al. (2022) proposed DroneTrace, an investigation strategy based on specific physical properties in drones (e.g., equivalent to the fingerprints and biometric traits in individuals), consisting of a low-cost millimeter-wave probe a software framework for extracting and processing parasitic actions, and a proprietary DNN-based methodology for analyzing and identifying drones. The authors also examined and predicted the behavior of parasitic electrical components during RF interrogation, which would be independent and difficult to forge.

## 3. Background

### 3.1. Drone architecture

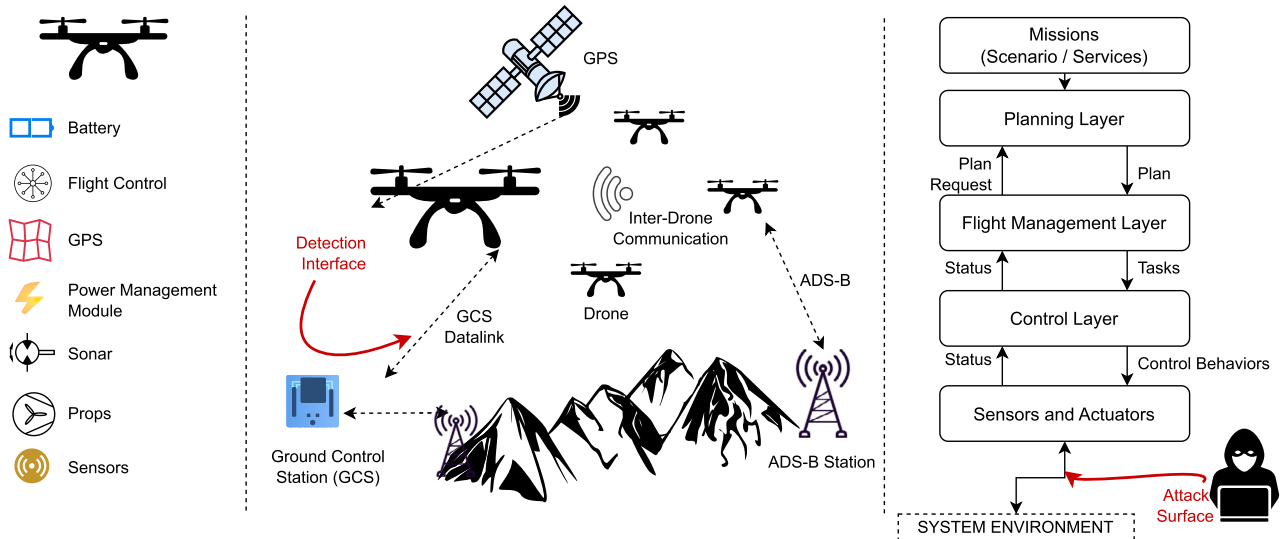
A drone system comprises three main components: the drone itself, the GCS, and the data communication link. The primary components of the drone include an aircraft, a power source, a flight controller, a precise navigation system, and a sensor system. Fig. 1 illustrates the architecture of a drone system and its key features.

#### Drone Craft

The drone craft is the primary mobile component of a drone system, which resembles a flying robot that can be remotely commanded or

**Table 1**  
Other related models and techniques based on drone RF fingerprinting.

| Reference                      | RF Signals | Mechanism                                          | Model | Key Contribution                                                                                                                            |
|--------------------------------|------------|----------------------------------------------------|-------|---------------------------------------------------------------------------------------------------------------------------------------------|
| Clemente et al. (2021)         |            | Chebychev moments based feature                    | ✓     | Chebychev moments' based feature for micro-Doppler Identification, with minimal computing cost, as well as orthogonality                    |
| Al-Emadi et al. (2019)         |            | CNN, RNN, CRNN                                     |       | collection of drone audio snippets                                                                                                          |
| Sciancalepore et al. (2019)    |            | J48, Random Forest, NN                             |       | passive detection of the drone                                                                                                              |
| Sciancalepore et al. (2020)    |            | Random Forest                                      | ✓     | network traffic analysis may be used efficiently for drone recognition                                                                      |
| Sheikh et al. (2019)           |            | Decision Tree, LR, KNN, DA                         |       | recommended collection of characteristics from a wireless network to be used as predictive variables in the categorization of flying drones |
| Xu et al. (2020)               | ✓          | SVM, KNN, STFT, PCA                                | ✓     | time-frequency-energy distribution properties of a single transient control, as well as video signals, are obtained for detection           |
| Elyousseph and Altamimi (2021) | ✓          | DFT                                                |       | 1D coefficient-based classifiers                                                                                                            |
| Bremnes et al. (2022)          | ✓          | CNN                                                |       | drone classification approach utilizing fixed boundary empirical wavelet sub-bands of RF signature                                          |
| Li et al. (2022)               | ✓          | DNN                                                | ✓     | strategy based on specific physical properties in drones consisting of a low-cost millimeter-wave probe                                     |
| Ender et al. (2021)            | ✓          | CNN                                                |       | radiofrequency time-series images and spectrograms were used to train the system                                                            |
| Ezuma et al. (2019a)           | ✓          | KNN, DA, SVM, NN                                   |       | attempted to address the detection issue and improve accuracy                                                                               |
| Ezuma et al. (2019b)           | ✓          | KNN, Random Forest, DA                             |       | address the identification and detection issues                                                                                             |
| Proposed Work                  | ✓          | CNN, KNN, Random Forest, XG-Boost, Ensemble Models | ✓     | DrIfTeR, drone RF fingerprinting model for detection, identification and classification of drones using 2 ensemble models.                  |



**Fig. 1.** Drone Architecture.

programmed with software-controlled flight plans. There are four main types of drones: Multi-Rotor Drones, Fixed-Wing Drones, Single-Rotor Drones, and Fixed-Wing Hybrid VTOL (vertical take-off and landing). Multi-rotor aircraft, such as tricopters, quadcopters, hexacopters, and octocopters, have many motors that provide vertical lift. In contrast, fixed-wing drones feature a rigid wing that functions like an airplane and only requires energy to move forward, making them more energy efficient. The flight controller is another essential component of the drone craft that reads sensor data, converts them into meaningful information, and either transmits them to the GCS or provides actuator control units directly with updated state information. The GCS communication interface is provided by the flight controller, as shown in Fig. 1.

#### Datalink

The datalink refers to the wireless connection established between the UAV and GCS, and serves to transmit control and data signals. The selection of the communication link is dependent on the UAV's operational range. UAV operations are categorized into two types: Line-Of-Sight (LOS) missions, which utilize direct radio waves to transmit and receive control signals, and BLOS missions, which depend on satellite systems or a relaying aircraft, which could also be a drone, to control the drone based on its distance from the GCS.

#### Ground Control Station (GCS)

GCS is the central structure that allows human operators to supervise and control drones during their missions. It provides a wireless connection to communicate with the drone, enabling the transmission of commands and the collection of real-time data. The size of a GCS may vary depending on the type and purpose of the drone. For tactical and strategic drones, a large self-contained facility with multiple workstations is often utilized as the GCS.

### 3.2. Drone communications

To enable UAV systems to perform military and civilian functions such as wildfire management, pollutant research, and disaster management, among others, they must communicate effectively with other entities in their network. The most common modes of communication between the drone and the base station are WiFi and radio. Although WiFi has a relatively low transmission range, radio signals can reach thousands of meters. A typical drone network comprises drones, GCS, navigation satellite systems, and Air Traffic Control (ATC) systems such as Automated Dependent Surveillance-Broadcast (ADS-B) systems, as shown in the Fig. 1. The network entities communicate through the following communication links:

- **GCS communication:** The datalink facilitates communication between unmanned aerial systems and GCS. The GCS utilizes wireless signals to establish communication with the UAVs, enabling it to supervise network traffic and control flight operations. However, since these communications rely on Wi-Fi/Bluetooth wireless technologies that have limited ranges, they are vulnerable to various types of attacks (Yaacoub et al., 2020).
- **ADS-B communication:** ADS-B (Automatic Dependent Surveillance-Broadcast) is a component of ATC system designed for aircraft to periodically broadcast navigation information using long-range RF signals. Operating on 1090 MHz, it transmits critical data such as altitude, GPS coordinates, and aircraft identification, enabling ground users and other aircraft to track and identify them. Recently, drones have begun incorporating ADS-B technology to enhance detect-and-avoid capabilities, ensuring safer airspace operations. However, traditional ADS-B systems are too bulky for smaller drones, necessitating the development of compact ADS-B modules for effective integration (Strohmeier et al., 2014; Park et al., 2021). It will improve drone tracking and identification while promoting safer skies.
- **Satellite and GPS Communication:** Using satellite communication, drones are adapting to meet the growing need for real-time data and

control. Satellite communication in drones enables long-range data transmission by connecting UAVs to satellite networks for reliable and real-time communication. By utilizing satellite links, drones can exchange critical information such as GPS coordinates, telemetry, and control commands, even in remote areas without ground-based infrastructure. Fixed-wing drones integrated with satellite systems are being explored as all-weather emergency communication platforms, overcoming the limitations of tethered drones (Zhou et al., 2019). Additionally, researchers are investigating the integration of 5G technology with SATCOM to improve network reliability, latency, and data efficiency (Singh et al., 2022). However, satellite communication systems often require high power and come at a significant cost, making their implementation more suitable for specialized applications.

- **Video transmission:** The RF data links that are commonly used for video transmission are 1.3 GHz, 2.4 GHz, and 5.8 GHz. (i) A range of over 40 miles and better penetration abilities can be achieved with the 3 GHz system, depending on the amount of power used. However, due to its low data rates, it produces poor video quality. (ii) The range of the 2.4 GHz system can be up to 15 miles, but using it for video transmission can cause interference as it is also used for control. (iii) The 5.8 GHz frequency is the most commonly used for video transmission due to its short wavelength and high data transfer rate capacity. It produces clear video quality compared to other options. However, its small wavelength limits its transmission distance to only 5 miles.

### 3.3. RF fingerprinting

RF Fingerprinting is a technique used to uniquely identify and identify radio devices based on their physical-layer signal characteristics during radio transmission. It leverages the fact that every wireless device due to manufacturing imperfections emits signals with slight, unique variations. These variations act like a "fingerprint" for that specific device. RF fingerprinting is considered a major enabler in object authentication and authorization to minimize the susceptibility of node forging due to its special ability to detect and identify transmitting devices (Jagannath et al., 2022). RF fingerprinting is a feasible and effective authentication and identification strategy to classify drones based on their transmitted signals (Shen et al., 2021). RF fingerprinting, like biometric fingerprinting, is unique and challenging to tamper with without significant effort.

#### RF Communication

Unmanned aerial systems are used for various military and civilian purposes, including pollutant research, wildfire management, disaster management, hurricane tracking, flood impact investigations, and illegal narcotic identification (Fig. 2).

WiFi or radio are the most common modes of communication between drones and base stations today. WiFi has a rather low transmission range, often a few hundred meters. The range of radio signals is thousands of meters. The drone comprises two primary components: the RC controller and the aircraft, both use a radio-frequency link to converse among themselves. The microcontroller may effectively operate the drone via control, data, and video signals to connect and transmit messages. The majority of microcontrollers operate on the 2.4 GHz frequency band and employ Frequency Hopping Spread Spectrum (FHSS) modulation. This technique has military applications because it can be employed for low detection probability to prevent the opponent from detecting the signals in the first instance. However, this feature makes detecting a drone more difficult. MAVLink protocol for drones establishes communication between the drone and the GCS, which helps to bidirectional communication. The protocol is generally used with ArduPilot and PX4 and offers extensive functionality in addition to control and tracking. The MAVlink protocol is composed of a series of signals that are sent between GCS and the drone. To ensure message integrity, the



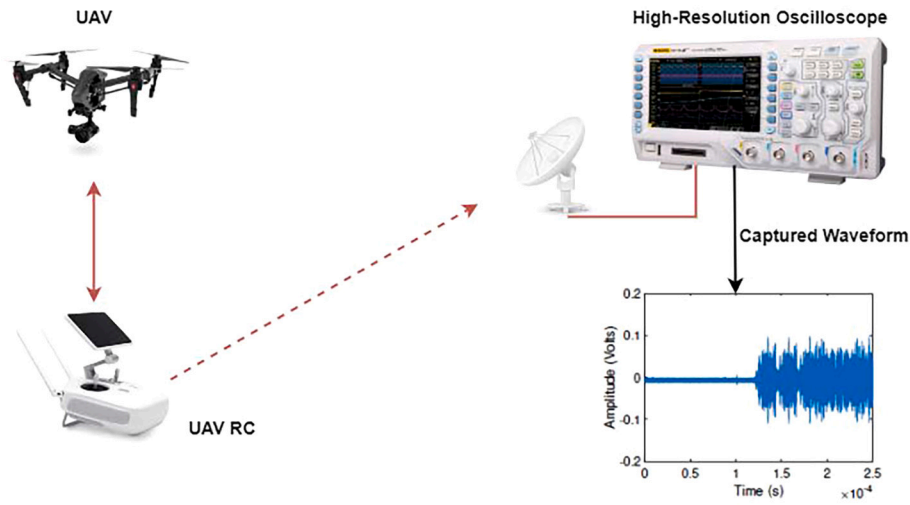


Fig. 2. Drone RF Communication.

protocol employs dual validation in the packet header (Koubâa et al., 2019).

Often groups and individuals use drone technology for specialized missions. It is necessary for security purposes to detect the presence of a drone in sensitive locations, identify the manufacturer details of the drone and the model details of the detected drone. As a result, it helps to determine whether there is a drone invasion in this area and to take the appropriate action. Different approaches for drone detection and identification have also been used for detection, identification, and classification challenges. Researchers have demonstrated the effectiveness of utilizing these methods to analyze RF signals in a wide range of practical scenarios (Al-Sa'd et al., 2019; Kosolyudhthasarn et al., 2018; Nguyen et al., 2016).

The drone architecture reveals multiple attack surfaces across both hardware and software components as depicted in Fig. 1. GPS and ADS-B links are vulnerable to spoofing and eavesdropping, enabling adversaries to misguide or monitor drones. The GCS Datalink can be exploited through jamming or hijacking attempts, disrupting control. Inter-drone communication and sensor-actuator layers are also exposed to signal injection and physical tampering. Additionally, attackers may target mission logic through the control and planning layers to alter drone behavior.

#### 4. Design of DrIfTeR

The architectural overview of DrIfTeR is illustrated in Fig. 3. The setup contains a UAV system, where a drone (UAV) communicates wirelessly with its remote controller (UAV RC). The RF signals transmitted during this communication are intercepted by an antenna and fed into a high-resolution oscilloscope, which captures the raw waveform of the transmission. This waveform, shown as a graph of signal amplitude over time. The captured signals are then used for pre-processing as described below. We have used Matlab R2024b software for signal processing and modeling.

##### 4.1. Pre processing

RF signal preprocessing in the DrIfTeR framework involves multiple stages designed to transform raw RF signals into informative feature vectors for classification. The steps are as follows:

**Wavelet Domain Extraction:** The received RF signal is first transformed from the time domain into the wavelet domain using a discrete wavelet transform (DWT). This transformation decomposes the signal into various frequency components localized in time, enabling a multi-resolution

analysis that is particularly suitable for non-stationary signals like RF emissions from drones. Given a discrete-time RF signal  $x(t)$ , we apply a Discrete Wavelet Transform (DWT) to convert it into the wavelet domain. The DWT decomposes the signal as:

$$x(t) \xrightarrow{\text{DWT}} \{A_j, D_j\}_{j=1}^L$$

where  $A_j$  and  $D_j$  represent the approximation and detail coefficients, respectively, at level  $j$ , and  $L$  is the total number of decomposition levels.

**3-Stage Wavelet Decomposition:** The wavelet decomposition is performed in multiple stages, typically referred to as levels. Each level of decomposition involves separating the signal into different frequency bands by using a high-pass filter (detail coefficients) and a low-pass filter (approximation coefficients). This process is iteratively applied in multiple stages to capture finer details of the signal's frequency components.

**Wavelet Coefficient Extraction:** At each stage of the wavelet decomposition, the resulting approximation and detail coefficients are obtained. These coefficients represent how the signal is distributed across different frequency bands and resolutions. The approximation coefficients capture the overall trend of the signal, while the detail coefficients represent the finer, high-frequency components. The output of this process is a structured, fixed-length feature vector for each RF signal sample.

In this work, we propose DrIfTeR, a drone detection model based on machine learning technology for identifying the existence of a drone in a specified location, manufacturer, and model details of the detected drone based on the acquired RF signals for that drone. In DrIfTeR, we use learning algorithms to implement the classification and detection phases. We evaluate the performance of CNN, k-NN, Random Forest (RFC), XG-Boost (XGB), and ensemble learning for classification and detection. The system is trained on labeled data and then evaluated on testing data. A 10-fold cross-validation method is employed for testing.

#### 5. Performance evaluation

This section introduces the datasets and assessment parameters. It then evaluates our suggested model against the following research questions.

- RQ1: Can DrIfTeR detect the presence of a drone with high accuracy? (*Drone Detection*)
- RQ2: Can DrIfTeR identify and classify the manufacturer of drones effectively? (*Drone Identification and Classification*)
- RQ3: Can DrIfTeR identify and classify the model of drone with high TPR and low FPR? (*Drone Identification and Classification*)

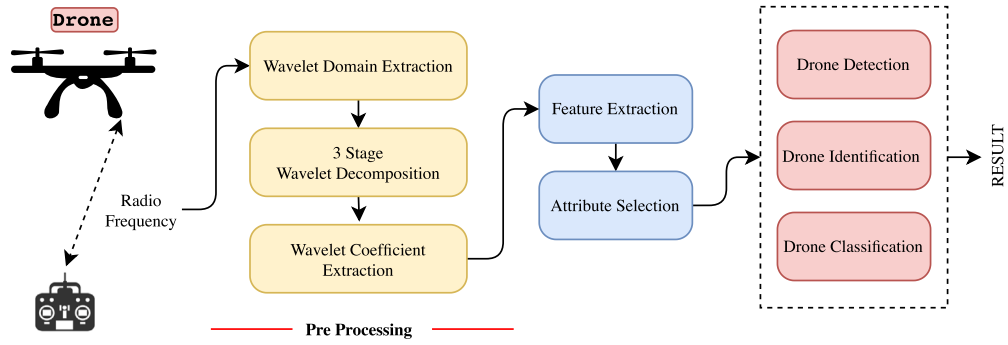


Fig. 3. Architecture of the proposed approach.

Table 2  
Dataset detail.

| Parameters            | Value     |
|-----------------------|-----------|
| No. of drone RCs      | 17        |
| No. of Signals/RC     | 1000      |
| No. of Samples/Signal | 5 Million |
| Time/Signal           | 0.25 ms   |
| Average Signal Size   | 7 MB      |
| Sampling Frequency    | 20 GSa/s  |
| Center Frequency      | 2.4 GHz   |
| Size of Dataset       | 124 GB    |
| Format of Dataset     | .mat      |

Table 3

Drone manufacturers and model used in dataset. (\*Two drones considered for this model).

| S.No. | Manufacturer | Model          |
|-------|--------------|----------------|
| 1     | DJI          | Inspire 1 Pro  |
| 2     |              | Matrice 100    |
| 3     |              | Matrice 600*   |
| 4     |              | Phantom 4 Pro* |
| 5     |              | Phantom 3      |
| 6     | Spektrum     | DX5e           |
| 7     |              | DX6e           |
| 8     |              | DX6i           |
| 9     |              | JR X9303       |
| 10    | Futaba       | T8FG           |
| 11    | FlySky       | FS-T6          |
| 12    | Graupner     | MC-32          |
| 13    | HobbyKing    | HK-T6A         |
| 14    | Jeti Duplex  | DC-16          |
| 15    | Turnigy      | 9X             |

### 5.1. Dataset and evaluation parameters

To answer the above-mentioned research questions, we evaluate the proposed approach against the benchmark dataset presented in Ezuma et al. (2020). This dataset encompasses RF signals originating from drone remote controllers (RCs) of diverse makes and models. Captured via a passive RF surveillance system comprising a high-frequency oscilloscope, directional grid antenna, and low-noise power amplifier, the dataset records signals transmitted in the 2.4 GHz band. Tables 2 and 3 provide dataset details and a list of the drones considered, respectively. A total of 17 drone RCs of 8 brands were used, 15 of which are unique. We divided the dataset into 2 classes, 9 classes, and 17 classes for answering respective research questions. For 2 class classifications, the nomenclature used is: Class 1 represents no-drone, and Class 2 represents drone presence. For the 9-class classification, the nomenclature used is: Class 1 to 9 represents no-drone, DJI, Spektrum, Futaba, Flysky, Hobbyking, Jeti Duplex, Turnigy, and Graupner manufacturers, respectively. For 17 class classification the nomenclature used is: Class 1 to 17 represents no-drone, Phantom 4 Pro I, Phantom 3, Matrice 100, Matrice 600 I, Phantom 4 Pro II, Matrice 600 II, JR X9303, DX5e, DX6e, DX6i, FS-T6, T8FG, MC-32, HK-T6A, DC-16, and 9X drone models respectively. Fig. 4 depicts the spectrum signals of the above three scenarios in the dataset (2, 9, and 17 classes) after preprocessing.

### 5.2. Methods for performance comparison

We used six machine learning algorithms for appropriate classification, namely: DNN (no of neurons = 256), k-NN (with  $k=13$ ), Random Forest (with the number of trees = 100), XGBoost (with number of estimators = 100), and two ensemble models. One ensemble model was using KNN and XGBoost, while the other was based on Random Forest and XGBoost. Samples for training and testing were used in a 90:10 ratio. We implement the aforementioned methods for training and testing. We employed 10-fold cross-validation for result validation. Table 4 lists the evaluation parameters used to measure performance.

#### RQ1: can DrIfTeR detect the presence of drones with high accuracy?

Drone detection problem deals with identifying the drone presence. We used 2 class dataset to answer the RQs. DrIfTeR's drone performance

with six different classifiers is evaluated. Table 6 and 7 shows the performance of DrIfTeR. Table 6 illustrates results using kNN, RF Classifier, XGB, Ensemble learning (KNN, XGB) and Ensemble learning (RFC and XGB). Table 7 illustrates the performance of DNN. Fig. 5 shows the classification performance using confusion matrix. It shows all drone and no-drone samples correctly classified. The numbers used for class representation (2-class) are listed in Table 5. Following are the conclusions based on the results:

- All classifiers function well with high accuracy (higher than 97%).
- DNN outperforms all other classifiers. KNN shows better performance than random forest, XGBoost and ensemble models.

⇒RQ1 Answer: DrIfTeR can effectively detect the presence of drones with high accuracy.

#### RQ2: can DrIfTeR identify the manufacturer of drone with high accuracy?

The issue of identifying drone manufacturers is a multi-class classification problem. For evaluating the performance of DrIfTeR for detecting drone manufacture we considered 9 classes. All six classifiers are evaluated against the dataset listed above and hard voting is considered to evaluate the performance with ensemble classifiers. The confusion matrix obtained using DNN is shown in Fig. 6. The confusion matrix shows the performance by the predicting classes of a model to the true classes of the data. The rows of the matrix represent the predicted classes, and the columns represent the true classes. The diagonal elements of the matrix represent the number of correct predictions, and the off-diagonal elements represent the number of incorrect predictions. In the figure, the confusion matrix shows that the model has good performance. The

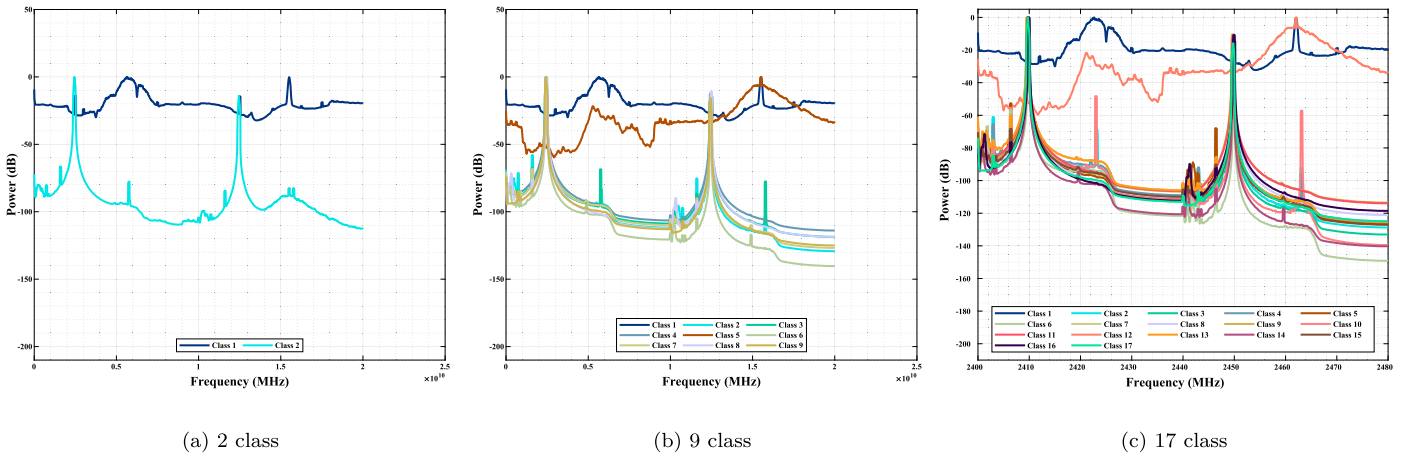


Fig. 4. Average spectra of RF activities.

Table 4

Drone detection and classification evaluation metrics.

| Term           | Abbreviation | Definition                                                                                                  |
|----------------|--------------|-------------------------------------------------------------------------------------------------------------|
| True Positive  | $TP$         | No. of samples correctly detected as drones or correctly classified into manufacturer/model $m$ .           |
| True Negative  | $TN$         | No. of samples correctly detected as no-drones or correctly not classified into manufacturer/model $m$ .    |
| False Positive | $FP$         | No. of samples incorrectly detected as drone or incorrectly classified into manufacturer/model $m$ .        |
| False Negative | $FN$         | No. of sample incorrectly detected as no-drones or incorrectly not classified into manufacturer/model $m$ . |
| Precision      | $p$          | $TP/(TP + FP)$                                                                                              |
| Recall         | $r$          | $TP/(TP + FN)$                                                                                              |
| F-measure      | $F_1$        | $2rp/(r + p)$                                                                                               |
| ROC Area       | $AUC$        | Area under ROC curve                                                                                        |
| Accuracy       | $Acc$        | Percentage of samples correctly detected or classified                                                      |

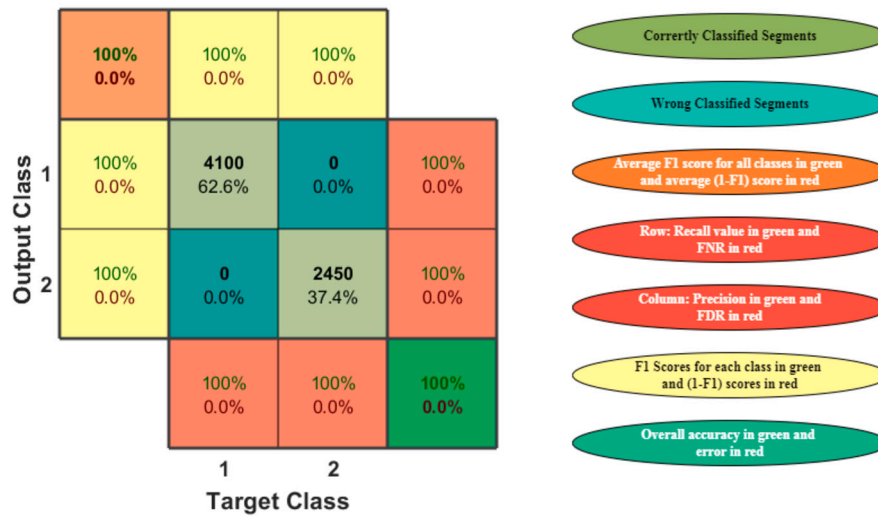


Fig. 5. Results: Drone presence detection by DriftER.

average F1 score is 99.3% and overall accuracy 99.6%. Most of the predictions are correct, and the model is able to distinguish between the different classes well. However, there are some errors. For example, the model predicted 30 samples of Graupner (class 9) as DJI (class 2) and 10 samples of DJI as Graupner. Similarly it misclassified few samples between class 3 (Spektrum), class 7 (Jeti Duplex) and class 8 (Turnigy) due to signal similarity. Fig. 7 shows box plot of accuracy (kNN, XGB and Ensemble model based on kNN-XGB). Table 6 and 7 illustrate performance results of DriftER for classification of drone man-

ufacturers problem. The numbers used for class representation (9-class) are listed in Table 5. The following conclusions are based on the above results:

- All classifiers perform well with high accuracy (higher than 96%).
- DNN outperformed with 99.6% accuracy and 99.3% F1 Score

⇒ **RQ2 Answer:** DriftER can identify and classify the drone manufacturer effectively.

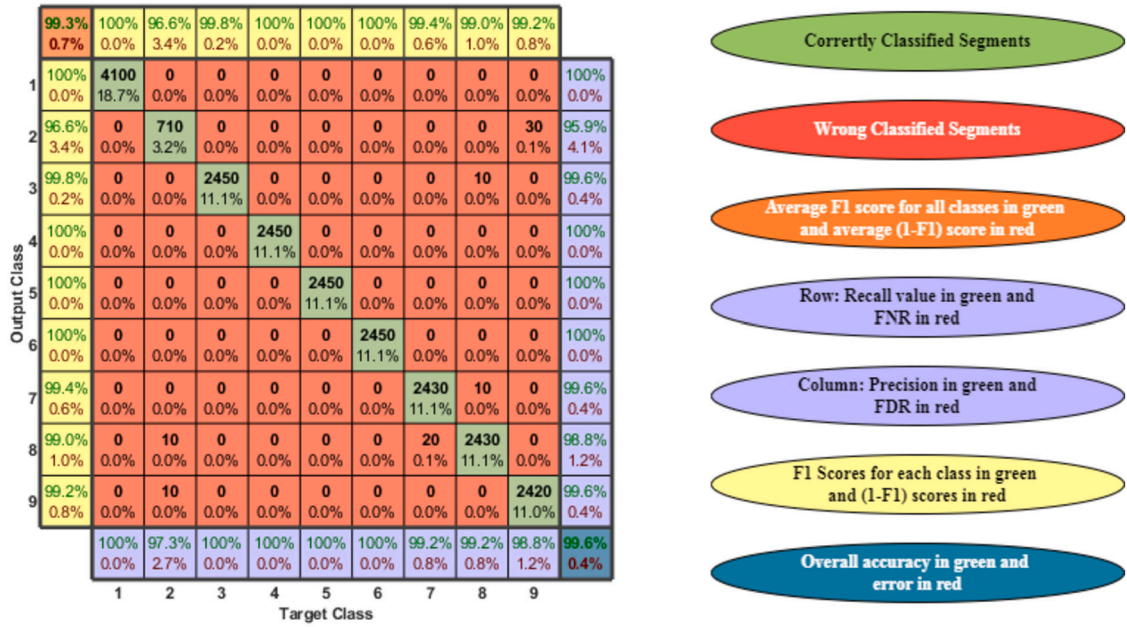


Fig. 6. Results: Drone manufacturer identification by DrifTER.

Table 5

Numerical values used to represent classes for 2 Class, 9 Class and 17 Class.

| 2 Class     | 9 Class        | 17 Class            |
|-------------|----------------|---------------------|
| 1: No Drone | 1: No Drone    | 1: No Drone         |
| 2: Drone    | 2: DJI         | 2: Phantom 4 Pro I  |
|             | 3: Spektrum    | 3: Phantom 3        |
|             | 4: Futaba      | 4: Matrice 100      |
|             | 5: Flysky      | 5: Matrice 600 I    |
|             | 6: Hobbyking   | 6: Phantom 4 Pro II |
|             | 7: Jeti Duplex | 7: Matrice 600 II   |
|             | 8: Turnigy     | 8: JR X9303         |
|             | 9: Graupner    | 9: DX5e             |
|             |                | 10: DX6e            |
|             |                | 11: DX6i            |
|             |                | 12: FS-T6           |
|             |                | 13: T8FG            |
|             |                | 14: MC-32           |
|             |                | 15: HK-T6A          |
|             |                | 16: DC-16           |
|             |                | 17: 9X              |

Table 6

Results: Comparative Analysis of performance for Drone presence detection, Manufacturer identification and model identification by DrifTER using KNN, RFC, XGB and Ensemble models.

| Classes  | Accuracy |        |        |           |           |
|----------|----------|--------|--------|-----------|-----------|
|          | KNN      | RFC    | XGB    | KNN + XGB | RFC + XGB |
| 2 Class  | 98.50%   | 97.30% | 97.30% | 96.27%    | 96.60%    |
| 9 Class  | 96.10%   | 97.00% | 96.20% | 96.30%    | 96.00%    |
| 17 Class | 96.19%   | 97.14% | 96.19% | 96.36%    | 96.60%    |

Table 7

Results: Comparative Analysis of performance for drone presence detection, manufacturer identification and model identification by DrifTER using DNN.

| Classes  | Accuracy | F1 Score | 1-F1 Score | Recall | Precision |
|----------|----------|----------|------------|--------|-----------|
| 2 Class  | 100%     | 100%     | 0%         | 1      | 1         |
| 9 Class  | 99.6%    | 99.3%    | 0.7%       | 0.880  | 0.992     |
| 17 Class | 95.9%    | 93.8%    | 6.2%       | 0.935  | 0.767     |

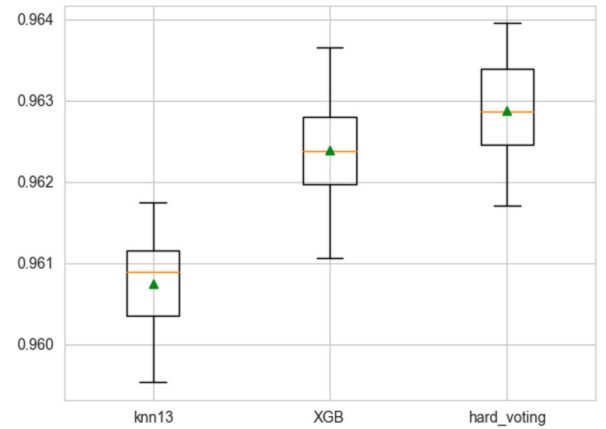


Fig. 7. Box Plot comparison of KNN, XGB and Hard Voting Ensemble (XGB+KNN) for drone manufacturer detection.

### 5.3. RQ3: can DrifTER identify and classify the drone model?

The issue of identifying a drone model is a classification problem. For evaluating the performance of DrifTER against this research question, we considered samples from 17 drone models. All six classifiers were evaluated against the dataset and hard voting is considered to evaluate the performance with ensemble classifiers. The Random Forest Classifier achieved an accuracy of 97.14%, outperforming the other classifiers. The confusion matrix from the DNN-based classifier, shown in Fig. 8, reveals a strong overall performance with an average F1-score of 93.8% and overall accuracy of 95.9%. The average F1 score is 93.8%, and the overall accuracy is 95.9%. Most predictions are accurate, demonstrating the model's ability to distinguish between different drone models effectively. However, non-trivial misclassifications were observed, particularly 240 samples of Class 15 (HK-T6A) being misclassified as Class 3 (Phantom 3). Similar confusion was noted between Class 6 (Phantom 4 Pro II), Class 7 (Matrice 600 II), and Class 10 (DX6e), which can be attributed to overlapping RF signature patterns due to shared hardware or transmission modules. Table 5 lists the numbers used for class representation (17-class). These misclassifications likely arise from similarities in modulation schemes, transmission frequencies, or antenna



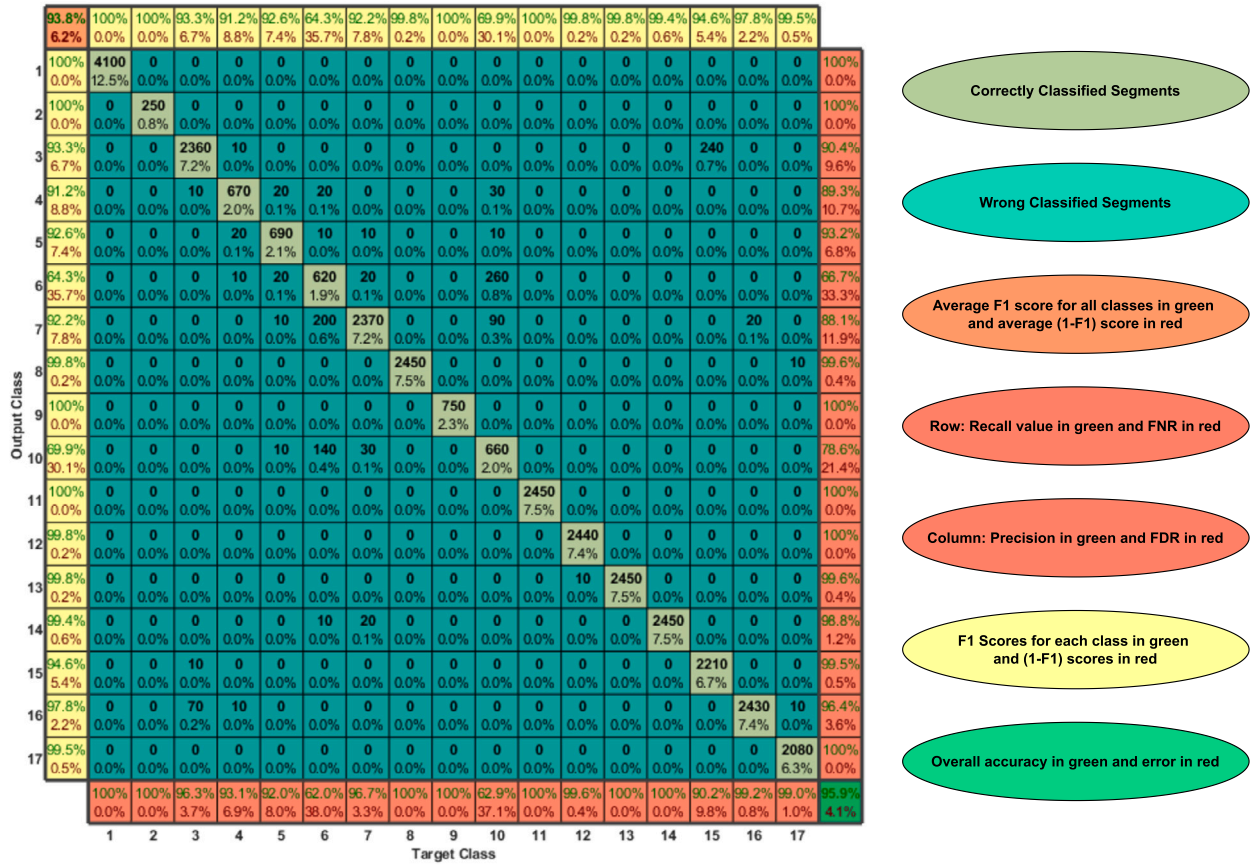


Fig. 8. Results: Drone model classification by DriftER.

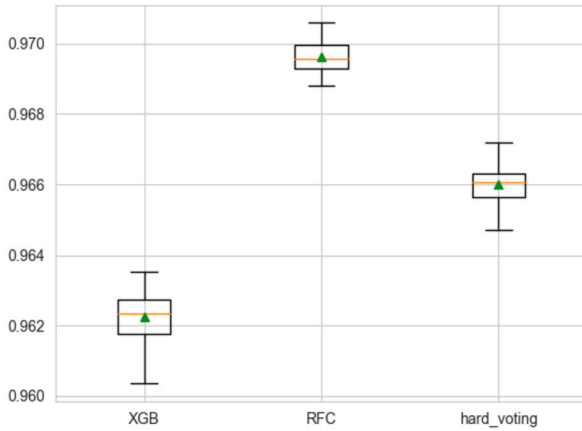


Fig. 9. Box Plot comparison of XGB, RFC and Hard Voting Ensemble (XGB+RFC) for drone model identification.

characteristics across certain drone models, which reduce the discriminative power of the extracted features. To mitigate this, future work may explore finer signal preprocessing techniques such as adaptive wavelet decomposition or time-frequency domain hybrid methods. Additionally, more robust feature extraction using deep learning-based representations (e.g., CNNs on spectrograms) or applying transfer learning from pre-trained RF models may improve the separation of closely related classes.

Fig. 9 shows the box plot comparison of accuracy (kNN, XGB and Ensemble model based on RFC-XGB). Table 6 and 7 illustrate performance results of DriftER for drone model identification problem. The following conclusions are based on the above results:

- All classifiers perform well with high accuracy
- DNN classifier is outperformed by traditional learning classifiers and ensemble model.

⇒ **RQ3 Answer:** DriftER is capable of identifying the drone model efficiently.

#### 5.4. Comparative analysis

Recent studies have increasingly focused on RF based methods for drone detection and classification. Medaiyese et al. compared various wavelet transform techniques for feature extraction, achieving an accuracy of 98.9% using wavelet scattering transform and SqueezeNet at a 10 dB signal-to-noise ratio (SNR) (Medaiyese et al., 2022). In Basak et al. (2021b), Basak et al. developed a novel RF dataset for drones and compared two-stage and combined detection-classification frameworks, finding that YOLO outperformed Goodness-of-Fit spectrum sensing in multi-signal scenarios. Nguyen et al. (2016) explored both active tracking and passive listening approaches using software-defined radio platforms, demonstrating the viability of RF-based drone detection. Garvanov et al. (2023) proposed a detection method based on RF emissions from drone-controller communication.

To evaluate the performance of our proposed model, we compared it with existing works that utilize the same benchmark dataset. We assessed our model against three studies Ender et al. (2021), Ezuma et al. (2019a), Ezuma et al. (2019b). In Ender et al. (2021), the authors implemented a CNN model using both RF time-series images and spectrograms for training. In Ezuma et al. (2019a), the authors aimed to enhance detection accuracy by using k-NN, discriminant analysis (DA), SVM, and neural networks. Finally, in Ezuma et al. (2019b), the authors employed KNN, random forest classifier (RFC), and DA methods to tackle identification and detection challenges using RF signals. A comparative analysis is provided in Table 8. The proposed approach performs well when com-

**Table 8**

A comparative analysis of proposed approach with existing works on Drone remote controller RF signal dataset (Ezuma et al., 2020). [DD: Drone Detection; MI: Drone Manufacturer Identification; Mol: Drone Model Identification].

| Paper                | Model      | DD | MI | Mol | Acc. (%)      |
|----------------------|------------|----|----|-----|---------------|
| Ender et al. (2021)  | CNN        |    |    | ✓   | 99.5          |
| Ezuma et al. (2019a) | kNN<br>SVM |    |    | ✓   | 96.30<br>> 95 |
| Ezuma et al. (2019b) | kNN        |    |    |     | 95.60         |
|                      | DA         |    |    |     | 92.77         |
|                      | SVM        |    |    | ✓   | 93.82         |
|                      | NN         |    |    |     | 92.88         |
|                      | RandF      |    |    |     | 96.32         |
| DrifTeR              | kNN        |    |    |     | 98.50         |
|                      | RandF      | ✓  |    |     | 97.30         |
|                      | NN         |    |    |     | 100           |
|                      | kNN        |    |    |     | 96.1          |
|                      | RandF      |    | ✓  |     | 97.00         |
|                      | NN         |    |    |     | 99.60         |
|                      | kNN        |    |    |     | 96.19         |
|                      | RandF      |    |    | ✓   | 97.14         |
|                      | NN         |    |    |     | 95.90         |
|                      |            |    |    |     |               |

pared with existing works in literature. The results also show that the benchmark dataset (Ezuma et al., 2020) is suitable for drone detection and model identification tasks.

## 6. Conclusion and future work

In this work, we proposed DrifTeR, a model for detection, identification and classification of different drones using RF signals. We used six classifiers comprising four individual classifiers and two ensemble models. The model was implemented and tested against benchmark dataset. Answers from RQs show that the proposed model is capable of detecting and identifying the drone, its make and model respectively with high accuracy.

RF-based methods have proven effective in detecting drones by analyzing signals emitted during communication between the drone and its controller. Future directions for drone detection involve several key areas of development. Firstly, there is a need to further explore and refine the active tracking and passive listening for detecting and characterizing drones using RF signals. Addressing the challenges identified in preliminary results will be crucial to enhancing the robustness and reliability of these systems. Additionally, a focus on creating automated and cost-effective drone detection solutions is essential. This includes developing hybrid systems tailored to specific scenarios, which account for detection range, environmental conditions, and drone identification without relying on the assumption that a drone has already been detected. Finally, expanding the datasets used for each detection approach will be vital. Larger and more comprehensive datasets will help validate performance, overcome limitations, and improve the overall accuracy of drone detection systems.

## CRedit authorship contribution statement

**Pankaj Choudhary:** Software, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Vikas Sihag:** Writing – review & editing, Visualization, Validation, Supervision, Project administration, Methodology, Conceptualization. **Gaurav Choudhary:** Writing – review & editing, Visualization, Validation, Supervision, Funding acquisition. **Nicola Dragoni:** Writing – review & editing, Validation, Supervision, Project administration, Funding acquisition.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Data availability

Data will be made available on request.

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